

From a technical and pure statistical point of view it is correct. However, from a real-life perspective, a margin of error may play a significant role. If contested by the affected party, such as loan applicants, chances of change in a decision are higher. Therefore these types of data points are of particular interest, and you should not ignore them.

The same applies when a human operator is supervising the AI system's output and can decide to override it. You must treat the human operator override as a special-case scenario, and feed it back to the training model. Each of these scenarios either highlights inadequacies in the base model or provides new situations that never existed before. Ignoring overrides can degrade the model performance over time.

5. Conflating correlation and causation

In statistics, we often say, "Correlation does not imply causation." It usually refers to the inability to legitimately deduce a cause and effect relationship between input variables and output. The resulting conclusion still may not be incorrect or false, but the failure to establish this relationship is often an indicator of the lurking problem.

On similar terms, the predictive power of your model does not necessarily imply that you have established an exact cause and effect relationship in your model. Your model may very well be conflating the correlation of input parameters and predicting output based on that.

You may think that "As long as it works, it shouldn't matter." However, the distinction matters since many machine learning algorithms pick upon parameters simply because there is a high correlation. Determining causality based on correlations can be very tricky and can potentially lead to contradictory conclusions. It would be much better to be able to prove that there truly is a causal relationship.

However, these days, developers and data scientists are merely relying on statistical patterns. Many of them fail to recognise that these patterns are only correlations amongst vast amounts of data rather than causative truths or natural laws, which govern the real world.

So, how do you deal with conflation?

Try this—during initial training and model building, soon after you find a correlation, don't conclude too quickly. Take time to find other underlying factors, find the hidden factors, and verify if these are correct and then only conclude.

The basis for trust

As technology is improving day by day, it is placing powerful tools in the hands of people who do not understand how these work. It is creating significant business as well as societal risks.

Developers and data scientists are increasingly getting detached from an understanding of intricacies of the tools they are using and systems they're creating.

The AI system means a black box is becoming a commonly accepted rhetoric, and the only sure-fire way to trust this black box is—training it meticulously and testing it rigorously! **END** 

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